

Distributivity and the Commensurability of Value-Definite Realism and Empirical Adequacy

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Abstract

When does an empirically adequate theory admit a value-definite completion — an assignment of simultaneous definite values to all observables? The answer depends on the algebraic structure of the observations. For a directed system of Boolean algebras carrying compatible charges, the Stone construction produces an unconditional measure on the dual space, and every observable admits a simultaneous definite value via ultrafilters. The passage to a probability measure on an underlying state space turns on σ -additivity of the charges, together with a Loomis–Sikorski realization condition on the choice of state space.

For orthomodular lattices — the algebraic framework for incompatible observations — the situation is structurally different. The McDonald–Bimbó dual space carries principal filters as distinguished structure, and the Kochen–Specker theorem blocks global value-definiteness. Probabilistic realism splits into two grades: Gleason-type results supply a σ -additive measure on the *lattice* (PR_{lat}) , but the passage to a measure on the *dual space* $(\text{PR}_{\text{dual}})$ fails — for the central case $L(H)$, no normal state extends to a charge on the dual, so even probabilistic realism in the dual-space sense is obstructed, and value-definite realism fails outright.

Distributivity guarantees value-definiteness; for the physically central case $L(H)$, $\dim H \geq 3$, non-distributivity blocks it. The state space admits a natural filtration $\mathcal{S} \supseteq \mathcal{S}_\sigma \supseteq \mathcal{S}_{\text{df}}$, whose three transitions — pasting, regularity, sharpness — are of different mathematical character. Distributivity collapses the filtration; non-distributivity can (but need not) open it.

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1 Introduction

A central question in the philosophy of science asks when an empirically successful theory can be completed to a realist one — one that assigns definite properties to an underlying reality, not merely adequate predictions to observable phenomena. Van Fraassen [30] argues that science need only aim at *empirical adequacy*: agreement with all observable phenomena. The scientific realist demands more — an underlying state space whose properties generate the observations.

This debate is usually conducted in philosophical terms. We give it a mathematical formulation.

Three nested positions, each a strengthening of the previous:

- (i) **Empirical adequacy (EA).** A measure on the dual space of the observation algebra that agrees with all observed charges. No commitment to an underlying state space.
- (ii) **Probabilistic realism (PR).** A probability measure on an underlying state space $(\Omega, \sigma(\mathcal{C}))$ that generates the observations. Requires descent from the dual-space measure.

- (iii) **Value-definite realism (VDR).** Every observable simultaneously has a definite value: a global dispersion-free state on the full observation algebra. On a Boolean algebra (and on $L(H)$) this coincides with a 2-valued homomorphism; on a general OML the two come apart (Remark below), and it is the dispersion-free state that is the operative notion throughout.

Both EA and PR as stated presuppose a genuine measure on the dual space: EA *is* such a measure, and PR descends from it. In the Boolean case this presupposition is free (the Stone construction supplies the dual-space measure unconditionally). In the OML case it is not, and PR accordingly refines into two grades — PR_{lat} , a measure on the lattice, and PR_{dual} , a measure on the dual space concentrating on the physical points — which coincide for Boolean algebras and separate for $L(H)$ (Theorem 3.1). Where the distinction matters we write the grade explicitly; an unqualified “PR” refers to the Boolean case, where the two agree. These are nested: VDR implies PR implies EA. Which implications reverse depends on the observation algebra:

Compatible observations. When all observations can be performed simultaneously, the observation algebra is a Boolean algebra — a complemented distributive lattice. Every Boolean algebra admits 2-valued homomorphisms (ultrafilters exist by Zorn’s lemma). In this setting, all three positions are available and mutually commensurable. The passage from EA to PR requires σ -additivity of the charges (and a realization condition on Ω , developed below); the further passage to VDR is free.

Incompatible observations. When some observations cannot be performed simultaneously — as in quantum mechanics, contextual experimental design, or distributed systems with interfering sensors¹ — the observation algebra is an orthomodular lattice (OML), a complemented but non-distributive lattice. The Kochen–Specker theorem [22] shows that for the prototypical case $L(H)$ with $\dim H \geq 3$, no 2-valued homomorphism exists: VDR is blocked. Probabilistic realism on the *lattice* remains available via Gleason’s theorem [13], but its *dual-space* form fails for normal states (no extension to a charge on the McDonald–Bimbó dual space $S_0(L(H))$, Definition 2.9). Empirical adequacy itself is hedged in the OML case: it is the orthogonally additive set function on the $\perp$ -stable clopens, which always exists, rather than a full measure on the dual space — promotion to such a measure is the extension step that fails for $L(H)$.

Distributivity guarantees value-definiteness; for $L(H)$ with $\dim H \geq 3$, its absence blocks it. The individual ingredients are known — not only the mathematical ones (Stone duality, the Kochen–Specker theorem, Gleason’s theorem, the Bub–Clifton uniqueness result, and the recent McDonald–Bimbó duality for orthomodular lattices [24]) but the philosophical reading as well: that value-definiteness is separable from realism and rests on the Boolean structure of jointly performable observations is Stairs [29]; that the realism question turns on distributivity is the Putnam [27]–Dummett dispute; and that the Boolean/non-Boolean event structure is primitive is Bub–Pitowsky [4]; and, in the von Neumann algebra setting, the separation between a state qua measure on the lattice and the Kochen–Specker absence of global points is the content of Döring’s spectral-presheaf analysis [8]. What we add is neither a new philosophical thesis nor a new theorem but a *bridge*: we carry this separation across the recent McDonald–Bimbó duality for orthomodular lattices [24] — a quantum-logic result with no prior contact with the philosophy-of-science literature — and read it against the van Fraassen empirical-adequacy/realism debate, organising EA, PR_{lat} , PR_{dual} , and VDR onto a single axis indexed by

¹The non-quantum examples require the incompatibility to be *operationally imposed*: the contexts must be genuinely non-co-realizable, with no joint probability space carrying all observations at once. This is the content of contextuality as failure of a global section (Abramsky–Brandenburger [1]; Fine’s theorem [12] for the joint-distribution criterion). Incompatibility cannot arise *intrinsically* from the dynamics of a single classical system: any finite family of classical observables on a common probability space jointly distributes (Kolmogorov), hence sits inside a Boolean algebra. Distributed-sensor contextuality is therefore a statement about non-co-realizable measurement settings, not about the dynamics of one underlying process.

distributivity. Within the philosophy-of-physics literature, and via this duality, the σ -additive passage to the dual space is not previously isolated. We begin with the Boolean case.²

Let $(\iota, \leq)$ be a directed set indexing a family of Boolean algebras $\{B_i\}_{i \in \iota}$ with injective homomorphisms $\varphi_{ij} : B_i \rightarrow B_j$ for $i \leq j$. Write $\mathcal{C} = \bigcup_i B_i$ for the direct limit — the *cylinder algebra*.

A *compatible family of normalised charges* is a collection $\{\ell_i\}$ with each $\ell_i : B_i \rightarrow [0, 1]$ finitely additive, $\ell_i(1) = 1$, and

$$\ell_j \circ \varphi_{ij} = \ell_i \quad (i \leq j).$$

The Stone space $\text{St}(\mathcal{C})$ is the space of ultrafilters on $\mathcal{C}$, compact and totally disconnected. Each $\omega \in \Omega$ determines a principal ultrafilter

$$\text{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}.$$

Any compatible family of normalised charges extends to a σ -additive Baire probability measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$ — unconditionally, with no assumption of σ -additivity. Each charge extends to a regular Baire measure on the compact, totally disconnected $\text{St}(B_i)$, where the clopens form a base and compactness makes finite additivity automatically σ -subadditive [28, 15]; compatibility then assembles these into $\hat{\mu}$ on the inverse limit $\text{St}(\mathcal{C})$ [6, Theorem 1]. This is the *empirically adequate* object: it agrees with all charges on cylinder events.

The measure $\hat{\mu}$ lives on $\text{St}(\mathcal{C})$, not on Ω . Descent to a probability measure P on $(\Omega, \sigma(\mathcal{C}))$ requires

$$\hat{\mu}(\text{pure}(\Omega)) = 1,$$

i.e., that $\hat{\mu}$ concentrates on the principal ultrafilters. Such concentration implies each ℓ_i is σ -additive; the converse holds when Ω realizes every countably-complete ultrafilter of $\mathcal{C}$, a Loomis–Sikorski condition that is automatic for countably generated systems but not in general.

Every Boolean algebra admits lattice ultrafilters — maximal proper filters — by Zorn’s lemma. On a Boolean algebra, maximality forces a dichotomy: for every a , either $a \in F$ or $a^c \in F$. The proof extends F to include a^c whenever $a \notin F$; the extension remains proper because $a^c \wedge f \neq \perp$ for all $f \in F$, which follows from distributivity. This dichotomy makes every lattice ultrafilter a 2-valued homomorphism, assigning each $a \in \mathcal{C}$ a definite value in $\{0, 1\}$. On a non-distributive OML, the extension argument fails, and maximal proper filters need not be 2-valued.

Moreover, any subset $S \subseteq \text{St}(\mathcal{C})$ that projects surjectively onto each $\text{St}(B_i)$ can serve as a realisation space: set $\Omega = S$ with evaluation maps given by the restriction maps. The algebra imposes no constraint on the choice of Ω .

Proposition 1.1 (Boolean commensurability). *For a directed system of Boolean algebras with compatible charges:*

- (a) *EA is unconditional: $\hat{\mu}$ on $\text{St}(\mathcal{C})$ always exists.*
- (b) *PR requires each ℓ_i to be σ -additive, and is equivalent to it under the Loomis–Sikorski realization condition on Ω (automatic for countably generated systems).*
- (c) *VDR holds unconditionally: $\text{St}(\mathcal{C})$ consists entirely of 2-valued homomorphisms.*

The obstruction at the EA–PR transition is σ -additivity of the charges, together with the Loomis–Sikorski realization condition on Ω (automatic for countably generated systems); both are conditions on the charges and the realization, not on the algebra.

²The Boolean construction is self-contained here; a fuller treatment of the directed-system Stone extension and the realization condition is developed in [23].

Proof. Part (a) is the Stone construction above [28, 6]. Part (b): descent to Ω requires $\hat{\mu}(\text{pure}(\Omega)) = 1$, which forces each ℓ_i σ -additive, and conversely under the realization condition (discussion above). Part (c): ultrafilters on a Boolean algebra are exactly the 2-valued homomorphisms; their existence follows from Zorn's lemma applied to the filter generated by any non-zero element. $\square$

2 Orthomodular observation algebras

Proposition 1.1 depends on distributivity at exactly one point: the existence of 2-valued homomorphisms. When observations are incompatible, this fails. The non-distributive structure need not be postulated: Gunji et al. [14] show that gluing Boolean algebras along shared boundaries via categorical pushout produces an orthomodular lattice, generically non-distributive. The OML is *forced* by the context structure.

Definition 2.1. An *ortholattice* is a bounded lattice $A = (A, \wedge, \vee, {}^\perp, 0, 1)$ with an orthocomplementation ${}^\perp : A \rightarrow A$ satisfying $a \wedge a^\perp = 0$, $a \vee a^\perp = 1$, $a^{\perp\perp} = a$, and $a \leq b \Rightarrow b^\perp \leq a^\perp$.

Definition 2.2. An *orthomodular lattice* (OML) is an ortholattice satisfying the *orthomodular law*:

$$a \leq b \implies b = a \vee (a^\perp \wedge b).$$

Every Boolean algebra is an OML (the orthomodular law follows from distributivity), but the converse fails. An OML is Boolean if and only if it is distributive [24, Proposition 2.3].

Example 2.3. The lattice $L(H)$ of closed linear subspaces of a separable Hilbert space H with $\dim H \geq 3$, ordered by inclusion with $U^\perp = \{x \in H : \langle x, y \rangle = 0 \ \forall y \in U\}$, is an orthomodular lattice that is not distributive [21, Chapter 1].

Example 2.4 (Incompatible measurements). Three binary measurements A, B, C , where (A, B) and (A, C) can be performed jointly but (B, C) cannot. The jointly measurable pairs generate Boolean subalgebras; the full observation algebra is orthomodular, not Boolean.

Two elements a, b in an OML *commute* if they generate a Boolean subalgebra. A maximal set of pairwise commuting elements forms a *Boolean context* — a maximal Boolean subalgebra of A .

Definition 2.5. A *state* on an OML A is a function $s : A \rightarrow [0, 1]$ with $s(1) = 1$ and

$$a \perp b \implies s(a \vee b) = s(a) + s(b),$$

where $a \perp b$ means $a \leq b^\perp$.

A state is additive on *orthogonal* pairs only — not on arbitrary joins. This is the OML analogue of a finitely additive charge on a Boolean algebra.

Definition 2.6. A state s is *dispersion-free* if $s(a) \in \{0, 1\}$ for all $a \in A$.

A dispersion-free state assigns a definite truth value to every proposition in A — the statistical/dispersion-free distinction in the standard quantum-logic sense [2]. Every 2-valued homomorphism is dispersion-free; the converse holds on Boolean algebras (where dispersion-free states are exactly the ultrafilters) and on $L(H)$, but *fails* on a general OML: a dispersion-free state need not be multiplicative across incompatible propositions. The minimal witness is MO_2 (the horizontal sum of two Boolean 2^2 blocks): assigning 1 to one atom of each block gives a dispersion-free state s , yet for the gap pair a, b with $a \wedge b = 0$ and $s(a) = s(b) = 1$ one has $s(a \wedge b) = 0 \neq 1 = s(a) \wedge s(b)$, so s is not a homomorphism. Throughout, the operative notion of value-definiteness is the dispersion-free state (equivalently, non-emptiness of $\mathcal{S}_{\text{df}}(A)$), not the strictly stronger 2-valued homomorphism.

Theorem 2.7 (Kochen–Specker [22]). *If $\dim H \geq 3$, there is no dispersion-free state on $L(H)$.*

For $\dim H = 2$, dispersion-free states exist; the obstruction is specific to $\dim H \geq 3$.

Theorem 2.8 (Gleason [13]). *If H is a separable Hilbert space with $\dim H \geq 3$, every σ -additive state on $L(H)$ is of the form $s(P) = \text{tr}(\rho P)$ for a unique density operator ρ on H .*

The analogue of Stone duality for OMLs is the McDonald–Bimbó duality [24].

Let A be an OML. A *filter* on A is a non-empty subset $x \subseteq A$ that is upward closed ($a \in x$ and $a \leq b$ implies $b \in x$) and closed under finite meets ($a, b \in x$ implies $a \wedge b \in x$). A filter is *proper* if $0 \notin x$; the unique *improper* filter is $\omega = A$.

For each $a \in A$, the *principal filter* generated by a is $\uparrow a = \{b \in A : a \leq b\}$. Write $F(A)$ for the collection of all filters on A , and $P(A) \subseteq F(A)$ for the subcollection of principal filters.

Definition 2.9 (McDonald–Bimbó [24]). Let A be an OML. The *dual space* of A is the ordered relational topological space

$$S_0(A) = (F(A), \subseteq, \perp_A, P(A), \mathcal{T}),$$

where:

- (i) $F(A)$ is the set of all filters on A ;
- (ii) $\subseteq$ is set inclusion;
- (iii) $x \perp_A y$ iff there exists $a \in A$ with $a \in x$ and $a^\perp \in y$;
- (iv) $P(A)$ is the collection of principal filters;
- (v) $\mathcal{T}$ is the topology generated by the subbasis $\{h(a) : a \in A\} \cup \{F(A) \setminus h(a) : a \in A\}$, where $h(a) = \{x \in F(A) : a \in x\}$.

Theorem 2.10 (McDonald–Bimbó [24]). *$S_0(A)$ is an orthomodular space (compact, totally disconnected). The category of OMLs and homomorphisms is dually equivalent to the category of orthomodular spaces and continuous weak p -morphisms. Moreover, $A \cong \text{CO}(S_0(A))^\dagger$, the OML of clopen $\perp$ -stable subsets of $S_0(A)$.*

A subset $U \subseteq S_0(A)$ is $\perp$ -stable if it is closed under the dual orthogonality: $y \in U$ and $x \perp_A y$ imply $x \notin U$. We write $\text{CO}(S_0(A))^\dagger$ for the sub-OML of clopen $\perp$ -stable subsets; its meet is intersection and its complement the dual orthocomplement, but its join is *not* set union.

The critical differences from Stone duality:

Feature	Boolean (Stone)	OML (McDonald–Bimbó)
Dual points	Ultrafilters	All filters
Distinguished subset	None	$P(A)$ (principal filters)
Clopens	Boolean algebra	OML
Orthogonality	Trivial	Non-trivial ($\perp_A$)
2-valued homomorphisms	Plentiful (Zorn)	May not exist (KS)

In Stone duality, the principal filters (i.e., the principal ultrafilters $\text{pure}(\omega)$ for $\omega \in \Omega$) are *invisible*: they play no distinguished role in the dual space. Which ultrafilters are principal depends on the choice of Ω .

In the McDonald–Bimbó duality, $P(A)$ is *part of the structure* of the dual space: the orthomodular frame conditions (Definition 2.9) explicitly reference $P(A)$. This is the algebraic formalisation: for non-distributive algebras, the distinction between realised and unrealised observation-patterns is constrained by the algebra. Distributivity is the condition under which this constraint vanishes.

2.1 Descent to the dual space and its obstruction

$S_0(A)$ is compact, so the topological machinery applies. But the algebraic input is weaker. A state s on A defines

$$\mu(h(a)) = s(a)$$

on $\text{CO}(S_0(A))^\dagger$.

In the Boolean case, $\text{Clop}(\text{St}(\mathcal{C}))$ is a Boolean algebra of sets, and s is finitely additive on it. The Carathéodory–Choksi extension machinery applies: compactness of $\text{St}(\mathcal{C})$ gives σ -subadditivity, and the extension to a Baire measure is automatic.

In the OML case, $\text{CO}(S_0(A))^\dagger$ is an OML of sets — closed under $\cap$ and $^\perp$ but *not* under arbitrary $\cup$. The state s is additive on orthogonal pairs only. Carathéodory does not apply: there is no Boolean algebra of sets on which s is finitely additive.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason’s theorem (Theorem 2.8) supplies a σ -additive measure *on the lattice* $L(H)$ for every σ -additive state: every such state arises from a density operator via the Born rule. This is a measure on A itself, and is *not* the same as a measure on the dual space $S_0(A)$ — a distinction the Boolean case obscures and the OML case forces. Indeed the passage to the dual space *fails* for $L(H)$: no normal (Gleason) state extends to a charge on the clopens of $S_0(L(H))$. This is a finitary non-extendability of the Kochen–Specker [22] and Pitowsky correlation-polytope [25] type, whose mechanism we sketch rather than prove in full.³ For general OMLs the extension problem is open.⁴

The conceptual separation between a measure on the lattice and the non-existence of a measure concentrated on points is, in the von Neumann algebra setting, already present in Döring’s analysis of the spectral presheaf [8]: states correspond to measures on the clopen subobjects of the presheaf Σ (PR_{lat} in our terms), while the absence of global sections of Σ is the Kochen–Specker obstruction (PR_{dual} failing). Our contribution is not this separation as such but its transposition: we phrase it through the McDonald–Bimbó duality for orthomodular lattices rather than the topos-theoretic presheaf, identify σ -additivity as the precise marker of the descent axis (Döring’s measures are finitely additive in general), and read the whole off against the van Fraassen empirical adequacy/realism debate, a framing absent from the topos literature.

Even when probabilistic descent succeeds, value-definite descent does not. The Bub–Clifton theorem [3] addresses what *can* be salvaged:

Theorem 2.11 (Bub–Clifton [3]; Halvorson–Clifton [16]). *Given a state ρ on $L(H)$ and a preferred observable R , there is a unique maximal sublattice $D(\rho, R) \subseteq L(H)$ such that:*

- (i) $D(\rho, R)$ is a Boolean algebra;
- (ii) $D(\rho, R)$ contains the spectral projections of R ;

³Within a single 2-dimensional subspace e , the lines p and their orthocomplements-within- e furnish, for every k , $2k$ pairwise meet-zero elements whose $\perp$ -stable clopen images are pairwise disjoint, while orthoadditivity gives $s(p) + s(e \ominus p) = s(e)$. A charge would then force $k s(e) \leq 1$ for all k , so $s(e) = 0$ for every finite-dimensional e — excluding every normal state. This is the 2-dimensional case of the Pitowsky correlation-polytope condition [25]; σ -completeness of the lattice is irrelevant, the obstruction being finitary. Only *singular* states ($s(e) = 0$ on all finite-dimensional e) escape, and whether one extends is open.

⁴The extension splits into two axes that the Boolean case fuses. The *extension axis* — passing from the orthogonally additive state on the OML of $\perp$ -stable clopens to a charge on the full Boolean algebra of clopens of $S_0(A)$ — is the classical Boolean extension problem [19]: since the representation map preserves meets, meet-zero elements map to disjoint sets, so the condition is $\sum_i s(a_i) \leq 1$ over pairwise-meet-zero families, a Pitowsky polytope condition [25] strictly stronger than orthoadditivity when the OML is non-distributive. It is not governed by σ -additivity, and in the finite case is a decidable linear program. The *descent axis* — whether the resulting measure concentrates on the physical points — is what σ -additivity governs. For $L(H)$, Gleason settles the lattice measure, but the extension axis itself fails (preceding footnote): Gleason does *not* deliver the dual-space measure.

(iii) The restriction of ρ to $D(\rho, R)$ is a mixture of dispersion-free states.

In the present vocabulary, condition (iii) says that $D(\rho, R)$ is the maximal Boolean subalgebra on which VDR is available — restating Halvorson and Clifton [16] in the EA/PR/VDR framework. It depends on R : different preferred observables yield different maximal Boolean subalgebras.

3 Distributivity as the commensurability condition

The status of the three positions depends on the algebra. In the OML case probabilistic realism splits into two grades that the Boolean case fuses: PR_{lat} , a σ -additive measure on the *lattice* A (Gleason, for $L(H)$), and PR_{dual} , a measure on the *dual* $S_0(A)$ concentrating on the physical points. We take PR_{dual} to be the *stronger* grade — a dual-space measure restricts, on the $\perp$ -stable clopens, to a finitely orthoadditive set function recovering s ; promoting that to a σ -additive lattice measure (PR_{lat}) is free in the Boolean case but only conditional in general (see (*)). The relationship the two grades certainly exhibit is a *witnessed separation*, not a proven nesting: for $L(H)$, PR_{lat} holds while PR_{dual} fails. The positions, ordered with the stronger to the right (a node marked $\dagger$ is not realised for $L(H)$):

$$\underbrace{\text{EA}}_{\perp\text{-stable set fn.}} \quad \Leftarrow \quad \underbrace{\text{PR}_{\text{lat}}}_{\text{Gleason } (L(H))} \quad \stackrel{(*)}{\Leftarrow} \quad \underbrace{\text{PR}_{\text{dual}}^\dagger}_{\text{fails for } L(H)} \quad \Leftarrow \quad \underbrace{\text{VDR}^\dagger}_{\text{KS-blocked (OML)}}$$

The implications run weak-from-strong: VDR (a global dispersion-free state) implies PR_{dual} (a dual-space measure); PR_{dual} implies PR_{lat} only conditionally (the restriction of $\hat{\mu}$ to the $\perp$ -stable clopens is finitely orthoadditive, but its σ -additivity on the lattice needs $\hat{\mu}$ to be null on countable-join defects — free in the Boolean case, conditional in general; see (*)); and PR_{lat} implies EA. For $L(H)$ the two strongest positions are not realised: VDR by Kochen–Specker, and PR_{dual} by the Kochen–Specker / Pitowsky non-extendability of §2.1 — the *upward* extension construction $\text{PR}_{\text{lat}} \rightarrow \text{PR}_{\text{dual}}$ (build a dual-space measure from the lattice measure) is obstructed, governed by distributivity rather than σ -additivity. Gleason delivers PR_{lat} but Gleason does not climb to PR_{dual} . The two grades coincide only in the Boolean case, where the extension is free and the ladder collapses to the single $\text{EA} \Leftarrow \text{PR}$ of the Boolean case. (*) The arrow $\text{PR}_{\text{lat}} \Leftarrow \text{PR}_{\text{dual}}$ is free in the Boolean case but holds only conditionally in general (it requires a σ -continuity hypothesis on the dual measure that the finitary McDonald–Bimbó duality does not supply); we use it only in the direction stated. EA in the OML case is the $\perp$ -stable clopen set function (the hedged reading below), not itself a dual-space measure.

Theorem 3.1 (Commensurability). *Let A be a complemented lattice.*

(a) *If A is distributive (Boolean), then EA, PR, and VDR are mutually commensurable:*

- *EA is unconditional (Stone construction).*
- *PR requires the charges to be σ -additive, and is equivalent to it under the Loomis–Sikorski realization condition on Ω . Here PR_{lat} and PR_{dual} coincide: the $\perp$ -stable extension step is free, so a lattice measure and a dual-space measure stand or fall together.*
- *VDR holds unconditionally (ultrafilters exist).*

Moreover, the choice of realisation space Ω is unconstrained by the algebra.

(b) *If A is non-distributive (OML with $A \cong L(H)$, $\dim H \geq 3$), then:*

- *EA is unconditional* in the $\perp$ -stable set-function sense.⁵
- *PR_{lat} holds* (Gleason, Theorem 2.8): a σ -additive measure on the lattice $L(H)$.
- *PR_{dual} fails for every normal state*: no charge on $S_0(L(H))$ extending s exists (Kochen–Specker / Pitowsky non-extendability, §2.1), so a fortiori none concentrating on $P(A)$. Equivalently, the extension step $PR_{lat} \rightarrow PR_{dual}$ is obstructed — the one transition the Boolean case leaves free.
- *VDR fails globally* (Kochen–Specker, Theorem 2.7).
- *VDR holds locally within each Boolean context* (Bub–Clifton, Theorem 2.11).

Proof. Part (a) is Proposition 1.1. For part (b): PR_{lat} follows from Gleason’s theorem (Theorem 2.8), which supplies a σ -additive measure on the lattice $L(H)$ for any σ -additive state. PR_{dual} fails for normal states by the Kochen–Specker / Pitowsky non-extendability of §2.1: no charge on the clopens of $S_0(L(H))$ extending s exists, so no such Baire measure on $S_0(L(H))$ and a fortiori none concentrating on $P(A)$. EA holds in the $\perp$ -stable set-function sense unconditionally — the assignment $\mu(h(a)) = s(a)$ on $CO(S_0(A))^\dagger$ always exists; note this does *not* go via PR_{dual} (which fails), since for $L(H)$ the genuine dual-space measure does not exist. VDR-failure is from Kochen–Specker; local VDR from Bub–Clifton. $\square$

Corollary 3.2. *Distributivity is sufficient for VDR. For the physically central case $A = L(H)$ with $\dim H \geq 3$, non-distributivity blocks VDR.*

Proof. Sufficiency: Theorem 3.1(a). Blockage: Theorem 3.1(b). $\square$

Remark 3.3. The dimensional restriction is necessary. For $\dim H = 2$, $L(\mathbb{C}^2)$ is non-distributive but admits dispersion-free states. More generally, horizontal sums of Boolean algebras and certain finite OMLs are non-distributive yet admit dispersion-free states [21, Chapter 4] — though, as MO_2 shows, these dispersion-free states need not be 2-valued homomorphisms. The sharp characterisation for general OMLs is: VDR is available iff A admits a dispersion-free state, which is a property of A , not of distributivity alone. Distributivity is sufficient (ultrafilters exist) but not necessary in full generality. For the physically and epistemologically central case $\dim H \geq 3$, it is the effective dividing line.

The containment $\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{df}(A)$ is standard in quantum logic [26, 21]. What we add is the identification of three transitions of different mathematical character, and the observation that distributivity controls which are trivial.

Definition 3.4 (Coherence filtration). Let A be a complemented lattice with a family of charges. The *state space* of A admits a descending filtration:

$$\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{df}(A),$$

where:

- $\mathcal{S}(A)$ is the set of states on A (contextual coherence);
- $\mathcal{S}_\sigma(A) \subseteq \mathcal{S}(A)$ is the subset of σ -additive states (probabilistic coherence);
- $\mathcal{S}_{df}(A) \subseteq \mathcal{S}_\sigma(A)$ is the subset of dispersion-free states (value-definite coherence).

Beneath the filtration sits a local condition: *consistency* requires only that each maximal Boolean subalgebra of A individually admits a compatible state.

⁵EA here is the orthogonally additive set function $\mu(h(a)) = s(a)$ on the $\perp$ -stable clopens, which always exists. It is *not* a genuine Baire measure on $S_0(A)$: promoting it to one is the extension step, which for $L(H)$ fails for every normal state (§2.1) and for general OMLs is open. The unconditional claim is thus about the clopen set function, not the dual-space measure.

Each inclusion can be strict. The three transitions have different mathematical character:

Transition	Character	Key	Obstruction
$\text{Consistency} \rightarrow \mathcal{S}(A)$	Pasting (topological)	Do local sections glue?	Contextuality
$\mathcal{S}(A) \rightarrow \mathcal{S}_\sigma(A)$	Regularity (analytic)	Does the state have good limits?	σ -additivity
$\mathcal{S}_\sigma(A) \rightarrow \mathcal{S}_{\text{df}}(A)$	Sharpness (algebraic)	Can probabilities collapse to $\{0, 1\}$?	Kochen–Specker

Distributivity collapses the filtration. For a Boolean algebra:

- *Pasting* is trivial: every local assignment extends to a global state (distributivity guarantees compatibility).
- *Sharpness* is free: ultrafilters exist (Zorn’s lemma), so $\mathcal{S}_{\text{df}}(A) \neq \emptyset$ unconditionally.
- The only non-trivial transition is *regularity*: σ -additivity of the charges (with the realization condition on Ω), as developed above.

For an OML with $\dim \geq 3$, every transition is non-trivial:

- *Pasting* is non-trivial: contextuality obstructions are possible.
- *Regularity* requires Gleason-type results. (This filtration grades *internal* states of A — i.e. PR_{lat} , the lattice measure, which Gleason supplies for $L(H)$; the further passage to a measure on the dual space $S_0(A)$, namely PR_{dual} — the extension step, free in the Boolean case but blocked by non-distributivity — is a separate question that for $L(H)$ *fails* for normal states and for general OMLs is open.)
- *Sharpness* is blocked: $\mathcal{S}_{\text{df}}(A) = \emptyset$ (Kochen–Specker).

Distributivity is sufficient to collapse the filtration. For $L(H)$ with $\dim \geq 3$, non-distributivity opens it fully.

Remark 3.5 (Relation to contextuality hierarchies). Abramsky and Brandenburger [1] introduced a hierarchy of contextuality — probabilistic, possibilistic, and strong — grading the *strength of the obstruction* to a global section of a presheaf of probability distributions. Our filtration is complementary: it grades *what the algebra can support*, not how badly a global section fails. Their axis measures obstruction strength; ours measures coherence demands.

4 Discussion

The standard philosophical debate between scientific realism and constructive empiricism treats the question as uniform: either theories aim at truth or they aim at empirical adequacy. The algebraic analysis above lets us recast that choice as *parametrised* by the observation algebra. For Boolean observations, the mathematics does not prefer one position over another. For non-Boolean observations ($\dim \geq 3$), VDR is algebraically impossible; the constructive empiricist position [30] is the generic one, and value-definite realism is the special case — the case in which the observation algebra happens to be distributive.

That value-definiteness is *separable* from realism — that the realist is not, qua realist, entitled to simultaneous definite values, and that what underwrites them is the Boolean (distributive) structure of jointly performable observations — is not new with us. It is the conclusion Stairs [29] reached directly: the requirement that all magnitudes carry classically definite values constrains the admissible structures for reasons independent of realism, turning on Boolean

representability rather than on realist commitment; his realism-without-value-definiteness position is, in the present vocabulary, the PR layer with $\mathcal{S}_{\text{df}}$ empty. The parallel link between realism and distributivity is the Putnam [27]–Dummett battleground (Putnam revises logic to keep value-definiteness; we instead read its loss off the algebra), and the primacy of the Boolean/non-Boolean event structure over any hidden dynamics is the Bub–Pitowsky [4] thesis. What the filtration adds to these is not a new philosophical position but an organising one: it locates EA, PR_{lat} , PR_{dual} , and VDR on a single axis indexed by distributivity, and separates the obstructions (pasting, regularity, sharpness) that the debate treats as one.

Several classical positions translate directly into this vocabulary, which is offered as a uniform restatement rather than a sharpening of any one of them. The EPR argument for completeness [11] amounts to the demand that all observables simultaneously possess definite values — i.e., VDR — which the Kochen–Specker theorem blocks algebraically for $\dim H \geq 3$. Bohr’s complementarity corresponds to the non-triviality of the pasting transition: incompatible contexts cannot be simultaneously realised. Bell’s beables [16] — the observables that *can* have definite values — are the elements of the maximal Boolean subalgebra $D(\rho, R)$ (Theorem 2.11), restating Bub–Clifton in this vocabulary. De Finetti’s operationalism is EA in the Boolean case; Pitowsky [25] develops the quantum analogue via Dutch book arguments on $L(H)$.

The analysis is not specific to quantum mechanics. Any collection of individually Boolean observation contexts — jointly performable measurements in behavioural science [10], jointly queryable attributes in machine learning [7], simultaneously administrable treatments in experimental design — produces an observation algebra via categorical gluing [14]. If the contexts are mutually compatible, the colimit is Boolean and VDR is available. If not, the colimit is orthomodular and VDR may be blocked. In principle this suggests a diagnostic: when a deterministic latent-state model fails, the failure may be algebraic — the observation algebra does not support VDR — rather than a consequence of noise or insufficient data. We state this only to mark its limits. By the equivalence used above⁶ “the failure is algebraic, not statistical” is the same diagnostic as “the failure is contextuality, not nuisance,” which the Contextuality-by-Default programme [10, 5] already operationalises on empirical data — and operationalises in the harder regime, since the sheaf/orthomodular formulation requires consistent connectedness (no marginal-selectivity violation), whereas real data routinely violates it and Contextuality-by-Default is built to handle that case. The contribution here is therefore not a new empirical test; it is the placement of that already-finitary diagnostic within the filtration, where the one obstruction it cannot see — the σ -additive, dual-space (PR_{dual}) layer, which is infinitary — becomes visible as a distinct transition.

Several directions remain open:

1. *The OML extension problem.* Does every state on a general OML extend to a σ -additive measure on $S_0(A)$, as in the Boolean case? For $L(H)$ the answer is *negative* for *every* state: Gleason supplies the lattice measure but no dual-space charge exists — the Kochen–Specker / Pitowsky non-extendability of §2.1 settles the normal states, and a finite family of infinite-dimensional, pairwise-meet-zero subspaces (a copy of MO_2 with infinite-dimensional atoms) settles the singular states. The general OML case remains open.
2. *Dynamical structure.* If the observation algebra carries a group action (stationarity, exchangeability), how does symmetry interact with the coherence hierarchy? The Boolean case is partially understood (de Finetti’s theorem for exchangeability); the OML case is unexplored.

⁶Non-distributivity of the colimit is equivalent to the non-existence of a joint distribution, i.e. to contextuality as failure of a global section [1, 12]; the categorical form of this equivalence, orthomodularity as the obstruction to a global section, is [14].

3. *Causal inference.* Holland’s fundamental problem of causal inference [18] — that both potential outcomes $Y(0)$ and $Y(1)$ cannot be observed for the same unit — is structurally analogous to Bohr complementarity. Whether partial identification constraints generate non-Boolean algebraic structure, and whether the commensurability theorem applies to counterfactual reasoning, is unexplored.
4. *Relation to the topos programme.* The Bohrification programme [20, 9, 17] encodes related structure more powerfully: the spectral presheaf over the poset of Boolean subalgebras captures VDR-failure as absence of global sections (Kochen–Specker), and local VDR within each Boolean context is built into the presheaf’s local sections. Our filtration operates at a coarser level — distinguishing EA, PR, and VDR as properties of the algebra’s state space rather than as sheaf-cohomological obstructions. The McDonald–Bimbó duality offers a complementary perspective: a single topological dual space (not a presheaf over contexts) in which principal filters are structurally distinguished. Whether the EA/PR/VDR vocabulary translates into the topos framework, and whether the filtration refines or merely simplifies the sheaf-theoretic picture, remains open.

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